An exact metric Γ-limit for the fat-Cantor nonlocal BV example

Candidate full theorem for the program in arXiv:2310.08882

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Status: candidate full theorem, likely valid. The nonlocal functionals in Example 6.3 of Lahti–Pinamonti–Zhou $\Gamma(L^1)$-converge, despite their failure to converge pointwise to one multiple of variation. For $1 < q < \infty$, their exact limit is

$$u \longmapsto 2^{1/q}\|Du\|_\mu.$$

More generally, if $d\mu = w\,dx$, $m \le w \le M$, and $w = m$ on an open dense set, then the limit is $2^{1/q}m^{1+1/q}|Du| = 2^{1/q}m^{1/q}\|Du\|_\mu$. The proof combines the Euclidean Γ-liminf with recovery sequences whose gradients are moved into the dense minimum-weight region.

1 The source problem and concrete target

Lahti, Pinamonti, and Zhou study nonlocal BV functionals on PI metric measure spaces. Their published Remark 1.9 proposes investigating Γ-convergence in that setting [1]. Their Example 6.3 is especially sharp: on $I = [0,1]$, let A be a fat Cantor set of Lebesgue measure $1/2$, put

$$w = 2\mathbf{1}_A + \mathbf{1}_{I\setminus A}, \qquad d\mu = w\,dx, \qquad \rho_i(x,y) = i\mathbf{1}_{\{|x-y|\le 1/i\}}, \tag{1}$$

and, for $1 < q < \infty$, define

$$\mathcal{F}_i^w(u) = \int_I \left[\int_I \frac{|u(x)-u(y)|^q}{|x-y|^q}\rho_i(x,y)w(x)\,dx\right]^{1/q} w(y)\,dy. \tag{2}$$

The paper exhibits Lipschitz functions with incompatible pointwise asymptotic constants. Its source also contains a commented concrete question asking whether these very functionals nevertheless Γ-converge to $a\|D\cdot\|_\mu$.

The published program and the two pages defining and analyzing the example are reproduced next.

If X supports a $(1,p)$–Poincaré inequality, $\Omega \subset X$ is a bounded pq-extension domain, and $f \in \widehat{N}^{1,pq}(\Omega)$, then

$$\limsup_{i\to\infty} \Phi_{p,q,i}(f,\Omega) \le C_2'' E_p(f,\Omega).$$

Here $C_1'' \le C_2''$ are constants that depend only on p, q, the doubling constant of the measure, the constants in the Poincaré inequality, and the constant C_ρ associated with the mollifiers.

Theorem 1.7. *Let $1 \le p < \infty$ and suppose $\varphi\colon [0,\infty) \to [0,\infty)$ satisfies assumptions (2.15)–(2.17), and suppose $\Omega \subset X$ is open and bounded.*

If μ is Ahlfors Q-regular with $1 < Q < \infty$ and $f \in L^p(\Omega)$, then

$$C_1''' E_p(f,\Omega) \le \limsup_{\delta\to 0} \Lambda_{p,\delta}(f,\Omega)$$

for some constant C_1''' depending only on p, the Ahlfors regularity constants, and the constant C_φ associated with the functional $\Lambda_{p,\delta}$.

If, on the other hand, μ is doubling, X supports a $(1,p)$–Poincaré inequality, and $f \in \widehat{N}^{1,1}(X) \cap \widehat{N}^{1,q}(X)$ for some $1 < q < \infty$ in the case $p = 1$, and and $f \in \widehat{N}^{1,p}(X)$ in the case $1 < p < \infty$, then

$$\limsup_{\delta\to 0} \Lambda_{p,\delta}(f,\Omega) \le C_2''' E_{f,p}(\Omega)$$

for some constant C_2''' depending only on p, the doubling constant, the constants in the Poincaré inequality, and the constant C_φ.

Remark 1.8. In general, one cannot have equality of the constants. In particular, in Theorem 1.6 one might hope for the equality $C_1'' = C_2''$. Brezis-Nguyen [7, Proposition 12] give an example where $\Phi_{1,q,i}(f,\Omega)$ defined with radial mollifiers satisfying (1.1) fail to converge to $K_{1,1}E_1(f,\Omega)$ for a Sobolev function $f \in W^{1,1}(\Omega)$ on $\Omega = (-1/2, 1/2)$. If f is further assumed to be in $W^{1,q}(\Omega)$ for some $q > 1$ on an open, bounded and smooth domain in $\mathbb{R}^n$, such convergence holds [7, Proposition 9]. However, we will give an explicit example (Example 6.3) showing that convergence to a constant times $E_1(f,\Omega)$ can fail even for a Lipschitz function on metric measure spaces.

Remark 1.9. It is worth pointing out that in some cases the convergence of the above functionals in the Euclidean space is not yet fully understood, for example $\Phi_{p,q,i}(f,\Omega)$ for a general $1 \le p < \infty$ [7, Section 3]. On the other hand, Γ-convergence of the functionals seems to be more robust and clear. In Euclidean spaces, the Γ-convergence of the functionals $\Phi_{1,q,i}(\cdot,\Omega)$, $\Psi_{1,i}(\cdot,\Omega)$, $\Lambda_{1,\delta}(\cdot,\Omega)$ is proved in [7, 10]. It would be interesting to investigate the Γ-convergence of such functionals in metric measure spaces as well.

The paper is organized as follows. We will review some definitions and introduce the conditions of the mollifiers ρ_i and the function φ in Section 2. Then we give some examples of ρ_i, φ and the related corollaries by applying such mollifiers in the main theorems in Section 3. We prove the lower bounds of our three main theorems in Section 4, and the upper bounds in Section 5. In Section 6 we give an example demonstrating that we do not generally have $C_1'' = C_2''$.

2. Notation and definitions

Throughout this paper, we work in a complete and connected metric measure space (X, d, μ) equipped with a metric d and a Borel regular outer measure μ satisfying a doubling property, meaning that there exists a constant $C_d \ge 1$ such that

$$0 < \mu(B(x, 2r)) \le C_d\mu(B(x,r)) < \infty$$

for every ball $B(x,r) := \{y \in X \colon d(y,x) < r\}$, with $x \in X$, $r > 0$. We assume that X consists of at least two points, that is, $\operatorname{diam} X > 0$. As a complete metric space equipped with $\mathbb{S}^{n-1} \subset \mathbb{R}^n$ is the unit sphere, $e \in \mathbb{S}^{n-1}$ is arbitrary, and $\mathcal{H}^{n-1}$ is the $n-1$-dimensional Hausdorff measure. Here the mollifiers ρ_i^* are required to satisfy (1.1). However, if f is only a BV function, it may happen that (6.2) fails, while a (different) limit exists, see [7, Proposition 10]. Many other counterexamples in the same spirit are given in e.g. [7, 10]. In metric spaces things can go even more seriously wrong, as we will now demonstrate.

Consider the real line equipped with the Euclidean metric and the one-dimensional Lebesgue measure $\mathcal{L}^1$. Consider the sequence of mollifiers

$$\rho_i(x,y) := \rho_i^*(|x-y|) := \frac{\chi_{[0,1/i]}(|x-y|)}{1/i}, \quad x,y \in \mathbb{R}, \quad i \in \mathbb{N}.$$

As noted before Corollary 3.7, these mollifiers satisfy assumptions (2.12) and (2.13). The mollifiers ρ_i^* obviously also satisfy (1.1). The example below, inspired by [27, Example 4.8], shows that even in a compact PI space and for a Lipschitz function f, the desired equality $C_1'' = C_2''$ in (6.1) may fail, and so in particular the Euclidean result (6.2) does not extend to metric spaces.

Example 6.3. Consider the space $X = [0,1]$, equipped with the Euclidean metric and a weighted measure μ that we will next define. First we construct a fat Cantor set A as follows. Let $A_0 := [0,1]$. Then in each step $i \in \mathbb{N}$, we remove from A_{i-1} the set D_i, which consists of 2^{i-1} open intervals of length 2^{-2i}, centered at the middle points of the intervals that make up A_{i-1}. We denote $L_i := \mathcal{L}^1(A_i)$, and we let $A = \bigcap_{i=1}^{\infty} A_i$. Then we have

$$L := \mathcal{L}^1(A) = \lim_{i\to\infty} L_i = 1/2.$$

Then define the weight

$$w := \begin{cases} 2 & \text{in } A, \\ 1 & \text{in } X \setminus A, \end{cases}$$

and equip the space X with the weighted Lebesgue measure $d\mu := w\, d\mathcal{L}^1$. Obviously the measure is doubling, and X supports a $(1,1)$–Poincaré inequality.

Let

$$g := 2\chi_A \quad \text{and} \quad g_i = \frac{1}{L_{i-1} - L_i}\chi_{D_i}, \quad i \in \mathbb{N}.$$

Then

$$\int_0^1 g(s)\, ds = \int_0^1 g_i(s)\, ds = 1 \quad \text{for all } i \in \mathbb{N}.$$

Next define the function

$$f(x) := \int_0^x g(s)\, ds, \quad x \in [0,1].$$

Now $f \in \mathrm{Lip}(X)$, since g is bounded. Approximate f with the functions

$$f_i(x) := \int_0^x g_i(s)\, ds, \quad x \in [0,1], \quad i \in \mathbb{N}.$$

Now also $f_i \in \mathrm{Lip}(X)$, and $f_i \to f$ uniformly. This can be seen as follows. Given $i \in \mathbb{N}$, the set A_i consists of 2^i intervals of length $L_i/2^i$. If I is one of these intervals, we have

$$2^{-i} = \int_I g(s)\, ds = \int_I g_{i+1}(s)\, ds,$$

and also

$$\int_{X\setminus A_i} g\, d\mathcal{L}^1 = 0 = \int_{X\setminus A_i} g_{i+1}\, d\mathcal{L}^1.$$

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Hence $f_{i+1} = f$ in $X \setminus A_i$, and elsewhere $|f_{i+1} - f|$ is at most 2^{-i}. In particular, $f_i \to f$ in $L^1(X)$ and so

$$\|Df\|(X) \le \lim_{i\to\infty} \int_0^1 g_i \, d\mu = \lim_{i\to\infty} \int_0^1 g_i \, d\mathcal{L}^1 = 1.$$

By Rademacher's theorem, for $\mathcal{L}^1$-a.e. $y \in A$, f is differentiable at y and so we have

$$\lim_{i\to\infty} \left(\int_X \frac{|f(x)-f(y)|^q}{|x-y|^q} \rho_i(x,y)\, d\mathcal{L}^1(x) \right)^{1/q} = (2|f'(y)|^q)^{1/q} = 2^{1/q}|f'(y)|.$$

Thus

$$\begin{aligned}
&\liminf_{i\to\infty} \int_X \left[\int_X \frac{|f(x)-f(y)|^q}{|x-y|^q} \rho_i(x,y)\, d\mu(x) \right]^{1/q} d\mu(y) \\
&\quad \ge 2 \liminf_{i\to\infty} \int_A \left[\int_X \frac{|f(x)-f(y)|^q}{|x-y|^q} \rho_i(x,y)\, d\mathcal{L}^1(x) \right]^{1/q} d\mathcal{L}^1(y) \\
&\quad \ge 2 \int_A \liminf_{i\to\infty} \left[\int_X \frac{|f(x)-f(y)|^q}{|x-y|^q} \rho_i(x,y)\, d\mathcal{L}^1(x) \right]^{1/q} d\mathcal{L}^1(y) \quad \text{by Fatou} \\
&\quad = 2^{1+1/q} \int_A |f'(y)|\, d\mathcal{L}^1(y) \\
&\quad = 2^{1+1/q}.
\end{aligned}$$

We conclude that

$$\liminf_{i\to\infty} \int_X \left[\int_X \frac{|f(x)-f(y)|^q}{|x-y|^q} \rho_i(x,y)\, d\mu(x) \right]^{1/q} d\mu(y) \ge 2^{1+1/q} \|Df\|(X). \tag{6.4}$$

On the other hand, consider any nonzero Lipschitz function f_0 supported in $(3/8, 5/8)$. For such a function, from (6.2) we have

$$\lim_{i\to\infty} \int_X \left[\int_X \frac{|f_0(x)-f_0(y)|^q}{|x-y|^q} \rho_i(x,y)\, d\mu(x) \right]^{1/q} d\mu(y) = 2^{1/q}\|Df_0\|(X) \in (0,\infty),$$

since both sides are equal to the classical quantities, that is, the quantities obtained when the measure μ is $\mathcal{L}^1$. This combined with (6.4) shows that we cannot have $C_1'' = C_2''$ in (6.1).

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2 A stronger weighted-interval theorem

The mechanism is clearer in the following general form. For a weight w, formula (2) retains its meaning with w in place of the fat-Cantor weight.

Theorem 1. *Let $I = [0,1]$, $1 < q < \infty$, and let $w : I \to (0,\infty)$ be measurable with*

$$0 < m \le w \le M < \infty \quad a.e. \tag{3}$$

Assume there is an open dense set $G \subset I$ on which $w = m$ almost everywhere. Set $d\mu = w\,dx$. Then, *with respect to* $L^1(I,\mu)$,

$$\mathcal{F}_i^w \xrightarrow{\Gamma} \mathcal{F}^w, \qquad \mathcal{F}^w(u) = \begin{cases} 2^{1/q} m^{1+1/q} |Du|(I), & u \in BV(I), \\ +\infty, & u \notin BV(I). \end{cases} \tag{4}$$

Moreover, the metric-measure variation obtained by relaxation against μ *satisfies*

$$\|Du\|_\mu(I) = m|Du|(I). \tag{5}$$

Consequently $\mathcal{F}^w(u) = 2^{1/q} m^{1/q} \|Du\|_\mu(I)$.

For the source's weight, $m = 1$ and $G = I \setminus A$ is open dense. Thus (4)–(5) answer the concrete question with

$$a = 2^{1/q}. \tag{6}$$

3 Moving variation into the minimum-weight set

We first record the recovery device responsible for both (5) and the Γ-limsup.

Lemma 2. *For every* $u \in BV(I)$, *there are* $g_k \in \mathrm{Lip}(I)$ *and compact sets* $K_k \subset G$ *such that*

$$g_k \to u \quad \text{in } L^1(I), \qquad g_k' = 0 \quad \text{a.e. on } I \setminus K_k, \qquad \int_I |g_k'|\, dx \longrightarrow |Du|(I). \tag{7}$$

Proof. Approximate u strictly in $BV(I)$ by step functions s_k with finitely many jumps:

$$s_k \to u \text{ in } L^1, \qquad \mathrm{TV}(s_k) \to |Du|(I).$$

For a fixed step function, move each jump by an arbitrarily small distance to a point of the dense open set G, preserving the order of the jumps. Around those new points choose pairwise disjoint closed intervals compactly contained in G. Replace every jump by the corresponding affine transition on its interval and leave the function constant elsewhere. The resulting Lipschitz function g has derivative supported in the union K of those intervals,

$$\int_I |g'|\, dx = \sum_{\text{jumps}} |\text{jump height}| = \mathrm{TV}(s),$$

and its L^1-distance from s can be made arbitrarily small. A diagonal choice proves (7). □

Proof of (5). The lower bound $w \geq m$ and the relaxation definition of metric-measure variation give

$$\|Du\|_\mu(I) \geq m|Du|(I).$$

For the sequence from Lemma 2, $w = m$ almost everywhere on the support of g_k', so

$$\int_I \mathrm{lip}\, g_k \, d\mu = m \int_I |g_k'|\, dx \longrightarrow m|Du|(I).$$

Taking the relaxation infimum gives the reverse inequality. □

4 The Γ-liminf

Let $\mathcal{F}_i^1$ denote (2) with Lebesgue measure, that is, with $w = 1$. From (3), pointwise in every measurable u,

$$\begin{aligned}\mathcal{F}_i^w(u) &= \int_I w(y) \left[i \int_{I\cap[y-1/i,y+1/i]} \frac{|u(x)-u(y)|^q}{|x-y|^q} w(x)\,dx \right]^{1/q} dy \\ &\geq m^{1+1/q} \mathcal{F}_i^1(u). \end{aligned} \tag{8}$$

The bounds in (3) make the $L^1(dx)$ and $L^1(\mu)$ topologies equivalent. Brezis–Nguyen's Euclidean theorem states that

$$\mathcal{F}_i^1 \xrightarrow{\Gamma(L^1)} \left[u \mapsto 2^{1/q} |Du|(I) \right] \tag{9}$$

for this radial mollifier [2, Proposition 15]. Therefore, whenever $u_i \to u$ in $L^1(\mu)$, (8)–(9) give

$$\liminf_{i\to\infty} \mathcal{F}_i^w(u_i) \geq 2^{1/q} m^{1+1/q} |Du|(I). \tag{10}$$

This also yields $+\infty$ when $u \notin BV(I)$.

5 The Γ-limsup

Fix $u \in BV(I)$ and take g_k, K_k from Lemma 2. Let

$$d_k = \operatorname{dist}(K_k, I \setminus G) > 0.$$

If $i > 2/d_k$ and the integrand in (2) is nonzero for a pair (x, y) with $|x - y| \leq 1/i$, then the segment between x and y meets K_k. Hence both points lie in G. Thus, for every fixed k and all sufficiently large i,

$$\mathcal{F}_i^w(g_k) = m^{1+1/q} \mathcal{F}_i^1(g_k). \tag{11}$$

The classical pointwise formula for Lipschitz functions gives

$$\lim_{i\to\infty} \mathcal{F}_i^1(g_k) = 2^{1/q} \int_I |g_k'|\,dx. \tag{12}$$

Choose increasing integers N_k so large that (11) holds and the difference between its left side and the limit in (12), after the factor $m^{1+1/q}$, is at most $1/k$ whenever $i \geq N_k$. Set $u_i = g_k$ for $N_k \leq i < N_{k+1}$. By (7),

$$u_i \to u \quad \text{in } L^1(\mu)$$

and

$$\limsup_{i\to\infty} \mathcal{F}_i^w(u_i) \leq 2^{1/q} m^{1+1/q} |Du|(I). \tag{13}$$

Equations (10) and (13) prove Theorem 1.

6 Why pointwise failure and Γ-convergence coexist

At density points of the high-weight Cantor set, a fixed smooth slope is sampled with both an inner and an outer copy of the large weight. This produces the source's anomalously large pointwise values. Γ-convergence instead permits recovery sequences. Lemma 2 moves each finite collection of transitions into nearby complementary intervals where the weight is minimal, and the diagonal choice makes the interaction scale much smaller than those intervals. This is precisely why the relaxed constant is the minimum weight rather than the pointwise high-weight density.

Thus the paper's counterexample to pointwise convergence is not a counterexample to Γ-convergence. It is an explicit positive instance of the metric-space program in Remark 1.9, and the dense-minimum theorem identifies the exact structural reason.

References

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